

SHI: Inspecting and Correcting Structured Hypotheses in Interactive Video Retrieval

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Abstract. Large language models are capable of interpreting complex video-search requests, but they may struggle to identify the correct event decomposition for complex narratives. Because an LLM-generated decomposition should not be treated as unquestionable, users should be able to inspect and correct it before it is propagated to retrieval. We present **SHI**, an interactive video retrieval system with two interaction stages: *pre-retrieval* and *post-retrieval*. In the pre-retrieval stage, the *Query Hypothesis* exposes the LLM-generated decomposition of a narrative into ordered events and allows users to revise it before search. In the post-retrieval stage, *EventTrail* exposes the temporal alignment of a candidate video and presents alternative event occurrences as complete ordered paths. User corrections are incorporated as temporal constraints and the alignment is re-decoded using the retained retrieval evidence.

Keywords: Interactive video retrieval · Video Browser Showdown · Temporal alignment · Human-in-the-loop · Multi-event search

1 Introduction

The Video Browser Showdown (VBS) benchmarks interactive video retrieval systems through live competition tasks over large video collections [18]. Multi-event Known-Item Search (KIS) is particularly demanding under competition conditions: queries describe a sequence of temporally ordered events that must first be interpreted as distinct semantic targets and subsequently aligned with valid temporal occurrences in candidate videos.

Large language models (LLMs) provide strong zero-shot capabilities for structuring natural-language descriptions into events, but automated decomposition remains imperfect, particularly when temporal dependencies must be inferred [20]. When an incorrect decomposition is passed directly to retrieval, errors in the inferred event structure can propagate to subsequent search stages. SHI therefore exposes this intermediate representation to the user before retrieval, allowing it to be inspected and corrected when necessary.

Prior VBS studies show that effective search depends on interaction beyond the initial ranking. A post-evaluation of VBS 2022 found that strong initial text-to-image retrieval alone did not yield the highest overall performance, with browsing and image-based

queries also contributing to search effectiveness [15]. Interaction logs from VBS 2026 likewise show repeated query reformulation and result inspection during search [8]. These observations motivate exposing intermediate retrieval decisions to the user rather than treating them as fixed outputs.

To address this challenge, we present **SHI**, an interactive video retrieval system with two stages of user interaction for multi-event KIS:

1. **Pre-retrieval Query Hypothesis:** SHI exposes the LLM-generated event decomposition before search. Users can revise the event structure, refine event descriptions, and attach image examples before executing retrieval.
2. **Post-retrieval EventTrail:** SHI exposes the temporal alignment of a selected candidate video as a complete ordered path. Users can keep the current occurrence, select an alternative, or reject an incorrect temporal region, after which the path is re-decoded under the updated constraints.

2 Related VBS Systems

VBS systems typically combine several retrieval modes with an interface for rapid result inspection. VISIONE provides textual, visual, and temporal search over a shared video index [1–3]. Vibro and diveXplore emphasize efficient browsing and visual exploration of large result sets [7, 16], while PraK builds its search interface on reusable video data services [12]. Recent versions of vitrivr likewise combine multimodal query formulation with a redesigned retrieval engine [5, 4]. SHI follows this system pattern: multimodal retrieval produces a ranked grid from which the operator can inspect candidates, open temporal context, and submit an answer.

Several VBS systems support interaction beyond an initial query. Text-query reformulation can use suggested concepts derived from early results [11]. Exquisitor combines conversational query revision with explicit relevance feedback on retrieved items [10, 17], and vitrivr has investigated interfaces that make retrieval decisions easier to inspect [6]. The *Query Hypothesis* in SHI applies this interactive principle to multi-event KIS: the event sequence produced from the description remains visible and editable before a search is executed.

Temporal queries have been expressed through structured conditions [19], contextualized multimodal stages [5], and sequence chains [9]. Interactive moment retrieval has also used feedback to guide navigation among candidate moments [13]. SHI uses an ordered-event decoder to obtain a complete path for each candidate video. *EventTrail* makes that path an interaction object: the operator can inspect alternative occurrences and apply event-level constraints before the path is decoded again.

3 System Overview

SHI separates offline corpus preparation from online interactive retrieval. Offline processing generates canonical frame metadata, multimodal evidence, and retrieval indexes. At runtime, these artifacts are loaded in read-only mode and shared by the KIS and AVS workflows. Figure 1 summarizes the retrieval pipeline used for multi-event KIS.

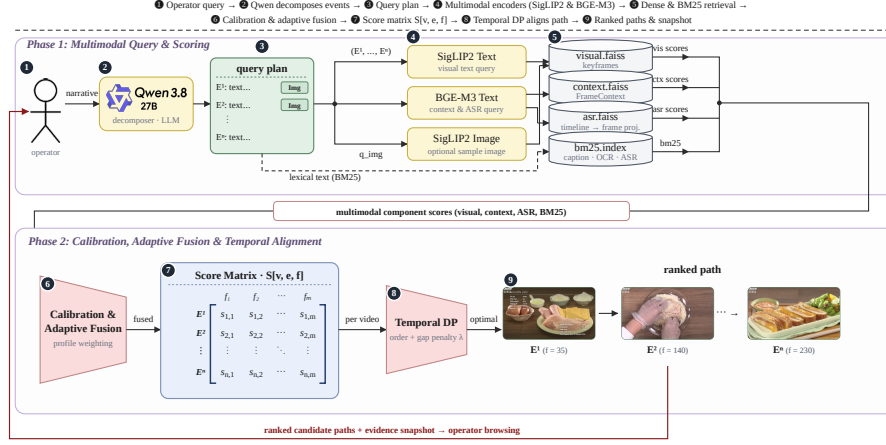


Fig. 1. Multi-event KIS retrieval in SHI. The Query Hypothesis provides an ordered sequence of events. Text events are searched against visual, FrameContext, ASR, and lexical evidence, while optional reference images contribute additional visual evidence. The component scores are calibrated and fused into an event–frame score matrix $S[v, e, f]$ for each candidate video. A dynamic-programming decoder then selects chronologically ordered event occurrences and ranks the resulting temporal paths.

3.1 Offline Evidence and Indexes

The VBS preparation profile targets the V3C collection [14]. Each sampled frame is stored with a canonical video identifier, frame identifier, frame index, and timestamp. SHI generates captions with Qwen3-VL-2B, extracts textual evidence with Florence-2, and detects objects with YOLOE. These modality-specific outputs are retained separately and also combined into a deterministic *FrameContext* representation for contextual text retrieval. Speech is stored independently as timestamped ASR segments.

SHI builds a SigLIP2 index over visual frames, a BGE-M3 index over FrameContext, and a BGE-M3 index over ASR segments. ASR scores are projected onto the canonical frame timeline when frame-level evidence is required. A separate BM25 index provides lexical retrieval over title, caption, OCR, and ASR fields. All enrichment and indexing steps are performed offline.

3.2 Multi-event KIS Retrieval

A multi-event KIS request is represented as an ordered sequence of events. Each event may contain text, reference images, or both. Text queries contribute dense visual, FrameContext, and ASR scores together with lexical BM25 evidence, while reference images contribute an additional visual score.

The modality scores are calibrated and fused only where the corresponding evidence is available. For each candidate video, this produces an event–frame score matrix with

one row per event and one column per canonical frame. A dynamic-programming decoder searches the matrix for complete paths that assign one frame occurrence to every event while preserving chronological order. Its objective combines event evidence with a penalty for large temporal gaps.

SHI retains multiple temporally distinct paths per video and ranks them across the collection. For each executed search, the event scores, decoded paths, event order, and decoder configuration of the returned videos are stored in an *Evidence Snapshot*. EventTrail operates on this snapshot: user corrections introduce temporal constraints and trigger re-decoding of the selected video without rerunning corpus retrieval.

3.3 AVS and Competition Integration

AVS follows a simpler retrieval path because it does not require an ordered event sequence. A text query searches the visual, FrameContext, and ASR representations, and SHI combines their ranked results using reciprocal-rank fusion. A coverage stage suppresses nearby results from the same video and interleaves candidates across videos before display.

Users can collect results from successive searches in a task-scoped basket and submit selected temporal answers to DRES. The backend maintains the competition session, validates the active task and answer type, and converts the selected video and temporal coordinates into the corresponding submission.

4 Interactive Hypothesis Correction

SHI provides two stages of user intervention for multi-event KIS. In the *pre-retrieval* stage, the *Query Hypothesis* exposes the LLM-generated interpretation of the input narrative before it is used for search. In the *post-retrieval* stage, *EventTrail* exposes the temporal alignment selected for a retrieved video and allows users to correct individual event occurrences without repeating corpus retrieval.

4.1 Pre-retrieval Query Hypothesis

For a multi-event request, the LLM produces an ordered event decomposition that is presented to the user as a *Query Hypothesis*. Each automatically generated event is grounded to a verbatim span of the original request, so the generated structure can be inspected together with its source text before retrieval.

Figure 2(a) shows the Query Hypothesis editor. Users can edit an event, split it, merge adjacent events, change the event order, or add a new event. Reference images can also be attached to individual events. These operations directly modify the event sequence that will be used by the downstream multimodal retrieval and temporal-alignment stages.

Changes are previewed before they affect the active hypothesis. The interface shows the current and proposed event sequences, after which the user may cancel or apply the modification. Applying a change creates a new query revision and invalidates results produced from the previous revision. Retrieval is executed only after the revised hypothesis is accepted by the user, and previous revisions can be restored through undo.

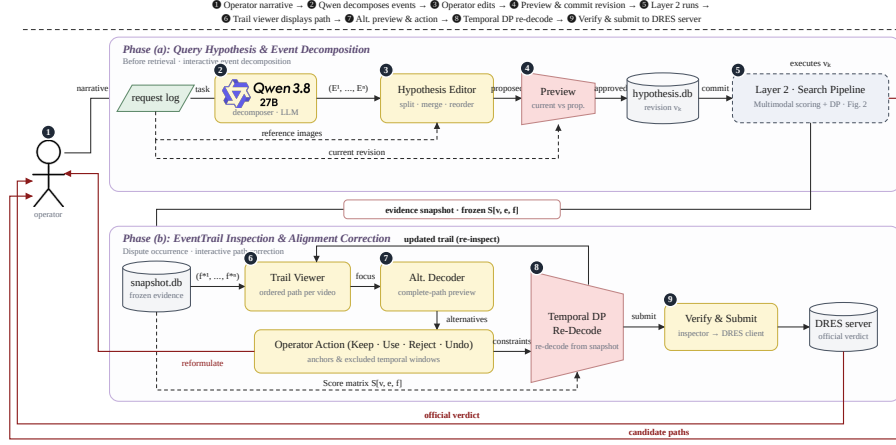


Fig. 2. The two interaction stages in SHI. **(a)** Pre-retrieval Query Hypothesis exposes the LLM-generated event decomposition and allows structural correction before search. **(b)** Post-retrieval EventTrail exposes the decoded temporal path and previews complete-path alternatives before a correction is committed.

4.2 Post-retrieval EventTrail

After retrieval, EventTrail exposes the current assignment of query events to occurrences within a selected video. Figure 3 presents this alignment as a complete ordered path, with each event associated with its selected frame and timestamp. The user can focus on any event and inspect temporally distinct alternatives under the current constraints.

An alternative is not presented as an isolated frame. For each candidate occurrence of the focused event, SHI decodes a complete chronologically valid path for the entire event sequence. This allows the user to preview how a local change would affect the complete alignment before committing it. If other event assignments must move to preserve chronological order, these changes are shown together with the previewed path.

Previewing does not modify the active path. The user can *Keep* the current occurrence, *Use* a previewed alternative, or *Reject* an incorrect temporal region. Keeping or using an occurrence introduces an anchor constraint for that event, whereas rejection introduces an exclusion constraint. SHI then re-decodes the complete path under the updated constraints using the evidence retained from the original search.

Because the path is decoded jointly over the complete event sequence, changing one event may also change the assignments of other events. EventTrail makes these changes visible in the preview, and previous corrections can be undone when necessary.

5 Future Work and Conclusion

SHI combines multimodal retrieval, temporal alignment, and task-aware submission in an interactive system for VBS 2027. For multi-event KIS, the *Query Hypothesis* exposes

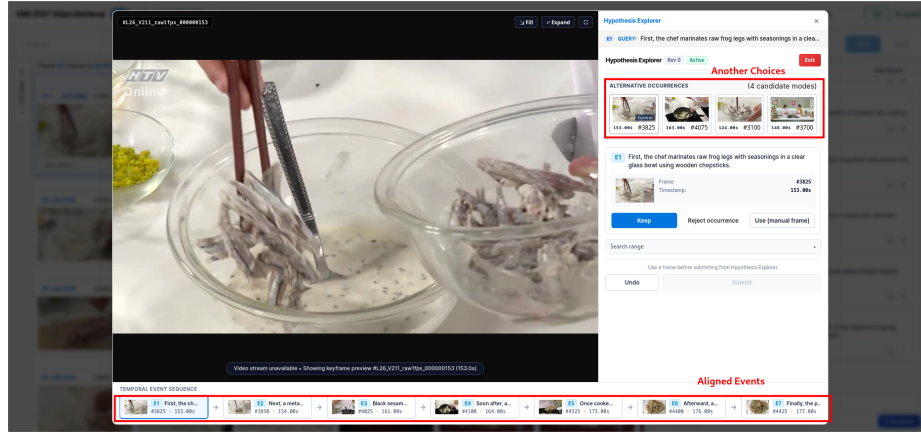


Fig. 3. EventTrail for a selected video. The bottom timeline shows the decoded event sequence, while the right panel provides alternative occurrences for the focused event. Users can *Keep* the current occurrence, *Use* an alternative, or *Reject* an incorrect temporal region before the path is re-decoded.

the event sequence used for retrieval, while *EventTrail* exposes the aligned path within a candidate video. Operators can therefore revise the query structure or constrain an incorrect temporal match from the search interface.

Future work will evaluate these interactions under competition conditions. We plan to compare the ranked-grid baseline, Query Hypothesis, and the complete SHI workflow using task success, search time, interaction count, and re-decoding latency. Separate ablations will test event decomposition, multimodal fusion, and anchor and exclusion feedback. These measurements will show when the additional controls assist search and when their interaction cost outweighs their benefit.

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